

Non-linear Recurrences

We explore some of the complexities of non-linear recurrences and solve one example of a non-linear recurrence for the Catalan numbers.

Class Session Goals:

- 1 Become familiar with multiple examples of the Catalan numbers.
- 2 Demonstrate using the generating function method to solve non-linear recurrence relations.

One homework exercise from last week

Find the first 5 terms of the Taylor expansion at $x = 0$ of $\sqrt{1 - 4x}$. By examining the pattern of coefficients find a_n where $\sqrt{1 - 4x} = \sum_{n=0}^{\infty} a_n x^n$.

Answer

$$\sqrt{1 - 4x} = 1 - 2x - 2x^2 - 4x^3 - 10x^4 - 28x^5 + \dots$$

or

$$\sqrt{1 - 4x} = 1 - \frac{1}{2^1 1!} 4^1 x - \frac{1 \cdot 3}{2^2 2!} 4^2 x^2 - \frac{1 \cdot 3 \cdot 5}{2^3 3!} 4^3 x^3 - \frac{1 \cdot 3 \cdot 5 \cdot 7}{2^4 4!} 4^4 x^4 - \frac{1 \cdot 3 \cdot 5 \cdot 7 \cdot 9}{2^5 5!} 4^5 x^5 + \dots$$

Another homework exercise from last week

Prove

$$\frac{1 \cdot 3 \cdot 5 \cdots (2n-1)}{2^n (n+1)!} 4^n = \frac{1}{n+1} \binom{2n}{n}.$$

Catalan Numbers Activity

In small groups, work on the “Catalan Numbers” worksheet. The goals of this worksheet are to:

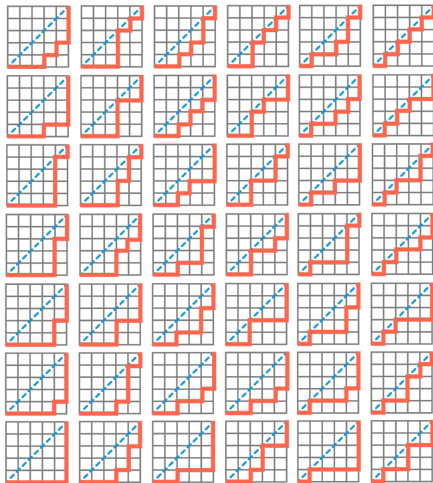
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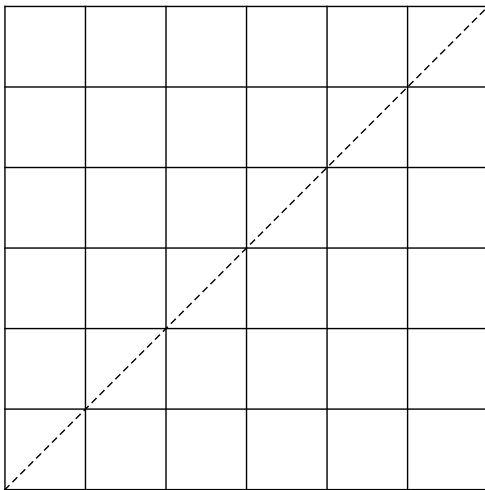
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Catalan numbers

1, 1, 2, 5, 14, 42, 132, 429, ... (A000108)

Create the Catalan Numbers Recurrence (n=6 case)



Catalan Numbers Recurrence

For $n = 6$, $C_6 = C_0C_5 + C_1C_4 + C_2C_3 + C_3C_2 + C_4C_1 + C_5C_0$ and in general we have

$$C_n = C_0C_{n-1} + C_1C_{n-2} + C_2C_{n-3} + \cdots + C_{n-1}C_0 \text{ or}$$

$$C_n = \sum_{i=0}^{n-1} C_i C_{n-1-i}$$

Closed form solution via the Generating Function method

Catalan numbers recurrence:

$$C_n = \sum_{i=0}^{n-1} C_i \cdot C_{n-1-i} \text{ with } C_0 = 1.$$

$$C_1 = C_0 \cdot C_0$$

$$C_2 = C_0 \cdot C_1 + C_1 \cdot C_0$$

$$C_3 = C_0 \cdot C_2 + C_1 \cdot C_1 + C_2 \cdot C_0$$

$$C_4 = C_0 \cdot C_3 + C_1 \cdot C_2 + C_2 \cdot C_1 + C_3 \cdot C_0$$

$$C_5 = C_0 \cdot C_4 + C_1 \cdot C_3 + C_2 \cdot C_2 + C_3 \cdot C_1 + C_4 \cdot C_0$$

Let $f(x) = \sum_{n=0}^{\infty} C_n x^n$ So let's try $f(x)^2$.

Use Generating function

$$f(x)^2 = (C_0 + C_1x + C_2x^2 + C_3x^3 + C_4x^4 + C_5x^5 + \dots) \quad (1)$$

$$\cdot (C_0 + C_1x + C_2x^2 + C_3x^3 + C_4x^4 + C_5x^5 + \dots) \quad (2)$$

So $f(x)^2 =$

C_0C_0+	C_0C_1x+	$C_0C_2x^2+$	$C_0C_3x^3+$	$C_0C_4x^4+$	$C_0C_5x^5+$	$\dots$
C_1C_0x+	$C_1C_1x^2+$	$C_1C_2x^3+$	$C_1C_3x^4+$	$C_1C_4x^5+$	$C_1C_5x^6+$	$\dots$
$C_2C_0x^2+$	$C_2C_1x^3+$	$C_2C_2x^4+$	$C_2C_3x^5+$	$C_2C_4x^6+$	$C_2C_5x^7+$	$\dots$
$C_3C_0x^3+$	$C_3C_1x^4+$	$C_3C_2x^5+$	$C_3C_3x^6+$	$C_3C_4x^7+$	$C_3C_5x^8+$	$\dots$
$C_4C_0x^4+$	$C_4C_1x^5+$	$C_4C_2x^6+$	$C_4C_3x^7+$	$C_4C_4x^8+$	$C_4C_5x^9+$	$\dots$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\ddots$

Use Generating function

So $f(x)^2 =$

$C_0 C_0 +$	$C_0 C_1 x +$	$C_0 C_2 x^2 +$	$C_0 C_3 x^3 +$	$C_0 C_4 x^4 +$	$C_0 C_5 x^5 +$	$\dots$
$C_1 C_0 x +$	$C_1 C_1 x^2 +$	$C_1 C_2 x^3 +$	$C_1 C_3 x^4 +$	$C_1 C_4 x^5 +$	$C_1 C_5 x^6 +$	$\dots$
$C_2 C_0 x^2 +$	$C_2 C_1 x^3 +$	$C_2 C_2 x^4 +$	$C_2 C_3 x^5 +$	$C_2 C_4 x^6 +$	$C_2 C_5 x^7 +$	$\dots$
$C_3 C_0 x^3 +$	$C_3 C_1 x^4 +$	$C_3 C_2 x^5 +$	$C_3 C_3 x^6 +$	$C_3 C_4 x^7 +$	$C_3 C_5 x^8 +$	$\dots$
$C_4 C_0 x^4 +$	$C_4 C_1 x^5 +$	$C_4 C_2 x^6 +$	$C_4 C_3 x^7 +$	$C_4 C_4 x^8 +$	$C_4 C_5 x^9 +$	$\dots$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\ddots$

Use Generating function

So $f(x)^2 =$

$C_0 C_0 +$	$C_0 C_1 x +$	$C_0 C_2 x^2 +$	$C_0 C_3 x^3 +$	$C_0 C_4 x^4 +$	$C_0 C_5 x^5 +$	$\dots$
$C_1 C_0 x +$	$C_1 C_1 x^2 +$	$C_1 C_2 x^3 +$	$C_1 C_3 x^4 +$	$C_1 C_4 x^5 +$	$C_1 C_5 x^6 +$	$\dots$
$C_2 C_0 x^2 +$	$C_2 C_1 x^3 +$	$C_2 C_2 x^4 +$	$C_2 C_3 x^5 +$	$C_2 C_4 x^6 +$	$C_2 C_5 x^7 +$	$\dots$
$C_3 C_0 x^3 +$	$C_3 C_1 x^4 +$	$C_3 C_2 x^5 +$	$C_3 C_3 x^6 +$	$C_3 C_4 x^7 +$	$C_3 C_5 x^8 +$	$\dots$
$C_4 C_0 x^4 +$	$C_4 C_1 x^5 +$	$C_4 C_2 x^6 +$	$C_4 C_3 x^7 +$	$C_4 C_4 x^8 +$	$C_4 C_5 x^9 +$	$\dots$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\ddots$

$$= C_1 + C_2 x + C_3 x^2 + C_4 x^3 + C_5 x^4 + \dots$$